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Numbering systems

I-13 Convert the following decimal numbers to octal and then directly to binary:

a) 653 ; b) 831.

Solution

a) $(653)_{10} = (1215)_8 = (1010001101)_2$

b) $(831)_{10} = (1477)_8 = (1100111111)_2$

I-14 Convert the following binary number into hexadecimal and octal:

a) 1101110110,111101

b) 110111, 0110111101

Solution

a) $(1101110110,111101)_2 = (1566.75)_8 = (376.F4)_{16}$

b) $(110111,0110111101)_2 = (67.3364)_8 = (37.6F4)_{16}$

I-15 Perform the following additions in binary:

a)	$\begin{array}{r} 1011.1101 \\ + 110.1011 \\ \hline \end{array}$	b)	$\begin{array}{r} 110111.01 \\ + 110011.11 \\ \hline \end{array}$
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Solution

a) 10010.1000

b) 1101011.00

I-16 Perform the following subtractions in binary:

a)	$\begin{array}{r} 101111101 \\ - 111111011 \\ \hline \end{array}$	b)	$\begin{array}{r} 110111.01 \\ - 110011.11 \\ \hline \end{array}$
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Solution

a) it gives a number without limit ...11110000010, nonsense result, subtraction not feasible in the classical way, later we'll see the two's complement method to apply in this case

b) 000011.10

I-17 Perform the following multiplications in binary:

$$\begin{array}{r} \text{a) } 11011,01 \\ \times 101,01 \\ \hline \end{array} \quad \begin{array}{r} \text{b) } 10111,11 \\ \times 110 \\ \hline \end{array}$$

Solution

a) 10001111.0001

b) 1000111.10

I-17 Perform the following multiplications in binary:

$$\begin{array}{r} \text{c) } 11001,011 \\ \times 1110 \\ \hline \end{array} \quad \begin{array}{r} \text{d) } 110111 \\ \times 10001 \\ \hline \end{array}$$

Solution

c) 101100011.010

d) 1110100111

I-18 Find 1) the one's complement; 2) the two's complement of the following binary numbers:

a) 1101110111 ; b) 1011011101 ;

Solution

C1 = the one's complement and C2 = two's complement

C1 of a) = 0010001000 C2 of a) = 0010001001

C1 of b) = 0100100010 C2 of b) = 0100100011

I-19 Perform the following subtractions using the two's complement:

$$\begin{array}{r} \text{a) } +9 \rightarrow 0 \ 1 \ 0 \ 0 \ 1 \\ -4 \rightarrow 1 \ 1 \ 1 \ 0 \ 0 \\ \hline \end{array} \quad \begin{array}{r} \text{b) } -9 \rightarrow 1 \ 0 \ 1 \ 1 \ 1 \\ +4 \rightarrow 0 \ 0 \ 1 \ 0 \ 0 \\ \hline \end{array} \quad \begin{array}{r} \text{c) } -9 \rightarrow 1 \ 0 \ 1 \ 1 \ 1 \\ -4 \rightarrow 1 \ 1 \ 1 \ 0 \ 0 \\ \hline \end{array}$$

Solution

Principle of calculation

Use of the two's complement in addition operations

In this explanation of the use of the two's complement in the calculations we consider

- the red bit, the sign bit
- the green bit, is a bit not considered because it is outside the number of bits we are working on.

+ 9 --> **0 1 0 0 1** --> 9 : positive number, it remains as it is with 0 assign bit **0 1 0 0 1**

- 4 --> **1 1 1 0 0** --> -4 : negative number then, take its two's complement with **1** as sign bit **1 1 1 0 0**

----- --> we perform the addition

+ 5 **1 0 0 1 0 1**

↑
not considered bit

we work on 4 bits, the fifth bit is not considered, the result is : **0 0 1 0 1**
the sign bit is 0, so the final result is = **0 0 1 0 1 = + 5**

If in the result we have **1** as a sign bit, then the final result will be **two's complement** of the found result, with of course (-) sign of the result.

hence the following operations:

+9 --> **0 1 0 0 1**

- 4 --> **1 1 1 0 0**

+5 --> **1 0 0 1 0 1**

↑
not considered bit

then

0 0 1 0 1 = + 5

- 9 --> **1 0 1 1 1**

+ 4 --> **0 0 1 0 0**

1 1 0 1 1

1 0 1 1 is the two's
complement of the result
so, it is the two's
complement of **5**

therefore Result = **- 5**

- 9 --> **1 0 1 1 1**

- 4 --> **1 1 1 0 0**

1 1 0 0 1 1

↑
not considered bit

0 0 1 1 = two's complement
of **13** then the result = **- 13**

I-20 Write in Gray code the following decimal numbers

- a) 27 ; b) 31 ; c) 192 d) 596.

Solution

a) 27 → **10110**

b) 31 → **10000**

c) 192 → **10100000**

d) 596 → **1101111110**

I-21 Write the decimal equivalent of the following BCD binary numbers:

- a) 0111 ; b) 0101 ; c) 1001 ; d) 1000

Solution

a) 0111 → **7**

b) 0101 → **5**

c) 1001 → **9**

d) 1000 → 8

I-22 Write in BCD the following decimal numbers:

a) 8732 ; b) 4149.

Solution

a) 8732 → 1000 0111 0011 0010

b) 4149 → 0100 0001 0100 1001

I-23 Give the code ASCII of 2; L ; t ; *.

Solution

2 → 0 1 1 0 0 1 0

F → 1 0 0 0 1 1 0

T → 1 0 1 0 1 0 0

* → 0 1 0 1 0 1 0

I-24 Give the symbol corresponding to 1001110 1001111 codes in ASCII.

Solution

1001110 1001111 → code ASCII de : N

I-25 Déterminer le nombre de fils requis pour envoyer le nombre 11010 :

- a) in series: two wires
- b) in parallel: five wires
- c) tell which transmission mode is the fastest: parallel
- d) which transmission mode is the most economical: serial

Solution

knowing that one voltage requires two wires

- a) one wire + ground
- b) Five wires + ground
- c) Parallel
- d) Serial

I-26 Code in reflected binary code (gray code)

a) the rows of the following table:



Solution

0000	0001	0010	0011	0100	0101	0110	0111	1000	1001

Good study with



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